

What is Variable Shadow ?

```
class Student
{
    static field
    non static field

    public void set(parameter)
    {
        local
    }
}
```

If class level fields and method level variables are having **exactly same name** then method level variable will **hide** class level fields **inside the method OR constructor OR block body**, this concept is known as Variable Shadow. [Method Level variables are having more priority than class level fields inside the method body]

```
VariableShadow.java
-----
package com.ravi.oop;

//ELC
public class VariableShadow
{
    public static void main(String[] args)
    {
        Student s1 = new Student();
        s1.accept(201);
    }
}

//BLC
class Student
{
    private int roll = 101;
    private String name = "Scott";
    public static String collegeName = "NIT";

    public void accept(int roll)
    {
        String name = "Alen";
        String collegeName = "VIT";

        IO.println(roll);
        IO.println(name);
        IO.println(collegeName);
    }
}
```

If I want to represent class level fields from the method body then what to do ?

* If we want to represent OR access class level fields from the method body then

- 1) For non static Field --> this keyword
- 2) For Static Field -> respective class name, as shown in the program below

```
package com.ravi.oop;

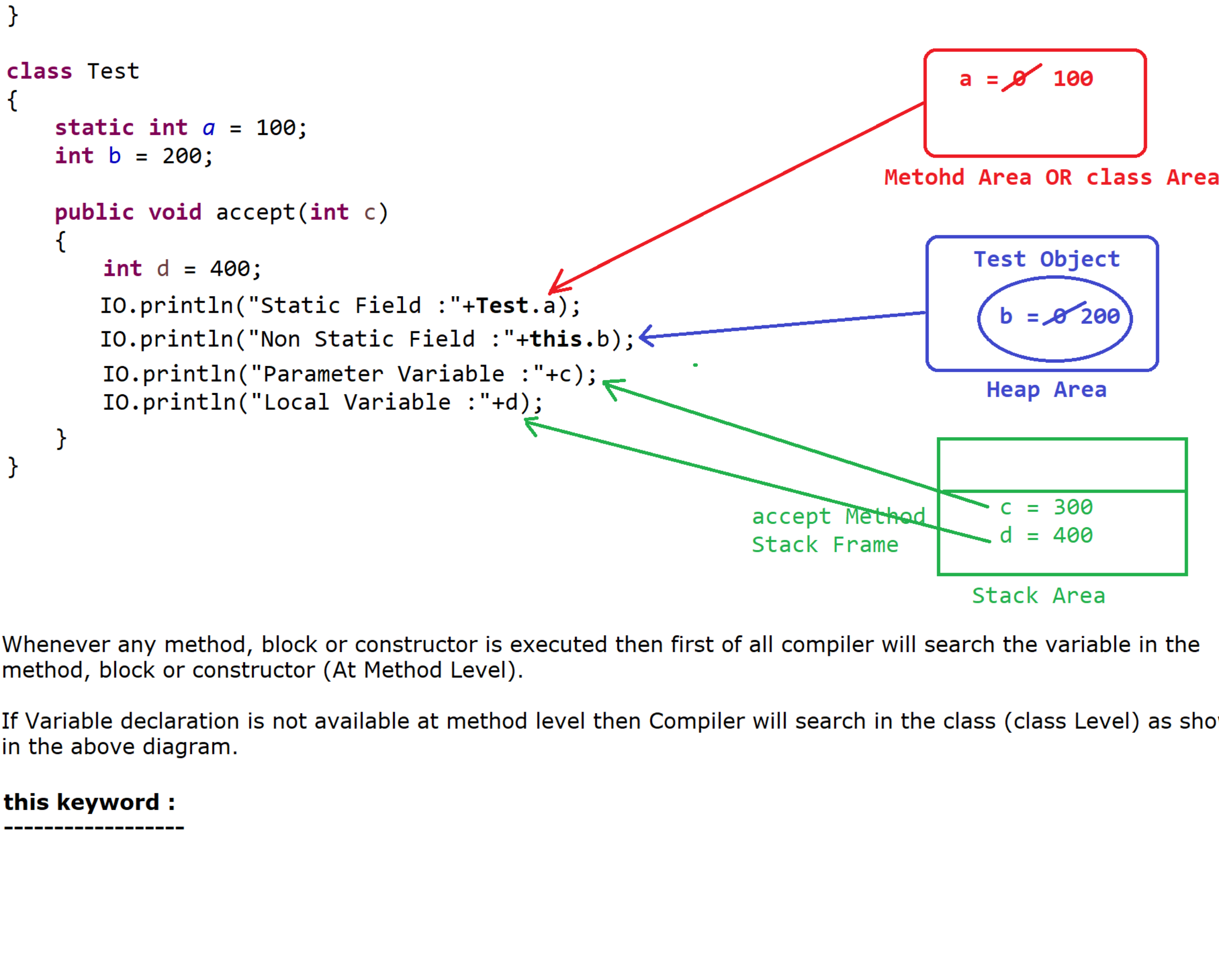
//ELC
public class VariableShadow
{
    public static void main(String[] args)
    {
        Student s1 = new Student();
        s1.accept(201);
    }
}

//BLC
class Student
{
    private int roll = 101;
    private String name = "Scott";
    public static String collegeName = "NIT";

    public void accept(int roll)
    {
        String name = "Alen";
        String collegeName = "VIT";

        IO.println(this.roll);
        IO.println(this.name);
        IO.println(Student.collegeName);
    }
}
```

What is local search algorithm ?



Whenever every method, block or constructor is executed then first of all compiler will search the variable in the method, block or constructor (At Method Level).

If Variable declaration is not available at method level then Compiler will search in the class (class Level) as shown in the above diagram.

this keyword :